NVIDIA GTX 1080 Ti FE

The NVIDIA GeForce GTX 1080 Ti effectively renders the GTX 1080 and GTX TITAN X worthless for ₹62,999

Announced at GDC 2017, the NVIDIA GeForce GTX 1080 Ti falls in line as yet another successful launch for NVIDIA. The company has been on a roll ever since Maxwell cards came out more than two years ago. And the lack of competition in this high end segment has been working out really well for NVIDIA which has enjoyed the lion's share of the enthusiast GPU market.

Price ₹62,999

A TITAN X with one less limb

The GTX 1080 Ti is based off the same GP102 that the NVIDIA TITAN X is based off of except for a few minor changes. And the GTX 1080 Ti is clocked at 1480 MHz which is 63 MHz higher than the Pascal based GTX TITAN. With GPU Boost, the clock speed goes up to 1582 MHz which is 51 MHz greater than the GTX TITAN.

The TITAN X had 12 GB of GDDR5X RAM clocked at 1251 MHz and the GTX 1080 Ti retains the use of GDDR5X but the clock speed has been increased to 1376 MHz which effectively takes the bandwidth from 10 Gbps to 11 Gbps. This is a significant increase in memory clock and when clubbed with the GPU clock, the overall processing power is more than the TITAN X. Yep, this little beast is more powerful than the TITAN X when it comes to gaming. To add perspective, the TITAN X had a single precision processing power of 10157 GFLOPs and the for GTX 1080 Ti that number is 10609 GFLOPS. Also, since NVIDIA has done away with 1 GDDR5X RAM chip it does away with a few ROPs as well. The TITAN X had 96 ROPs in the GPU so with 1 less chip, the ROP count goes down to 88.

DirectX 12 Ahoy!

Now that DirectX 12 has become really widespread thanks to Microsoft forcing it down everyone's throats, it only makes sense that NVIDIA should start devoting more resources to it. And this is exactly what they've done with the new driver update.

Build

The NVIDIA GeForce GTX 1080 Ti doesn't vary much from the previous Founders Edition cards released. Flipping the card around to its I/O plate shows a missing port! NVIDIA has done away with the DVI-D connector. Instead, they're shipping a DisplayPort++ to DVI-D adapter in the package. NVIDIA says this design decision allows for 2x the area for improved airflow.

Turns out, the new design does in fact result in better air flow as the temperature recovers faster than the previous design. There's a difference of 25.35 seconds so the cooling is 16.06 percent faster with the new design.

Performance

It should be noted that in none of the games are the 1080p FPS scores worth mentioning since they are all above 150. Rise of the Tomb Raider run really well on 1440p and every FPS score was above 100. On Ultra we managed to get 114.68 FPS and then we see the 4K benchmarks which finally brought the FPS scores down to double digits. On Ultra, we scored 63.8 FPS which is above the magical 60 FPS figure. In our compute tests, between the GTX 980 Ti and the GTX 1080 Ti, we recorded a 65% increment.

Verdict

NVIDIA has had a great year in 2016 and by the looks of it seems that 2017 is going to the same. The GTX 1080 Ti provides great value to the consumer, especially since it's nearly a TITAN X Pascal (even better in certain benchmarks) for a lot lower price tag.

One thing that stands out is that in all of our 4K benchmarks at Ultra settings, the GTX 1080 scored above the bare minimum of 60 FPS which means that you can now have the best 4K no-compromises experience with the NVIDIA GeForce GTX 1080 Ti. All you need is ₹62,999.

Mithun Mohandas

Interactive review

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Performance.............. 90
Build.............................75

Specifications

GPU: GP102; **CUDA Cores:** 3584; **TMUs:** 224; **ROPs:** 88; **Core clock:** 1480 MHz; **Boost clock:** 1582 MHz; **Memory Data rate:** 11 Gbps; **Memory:** 11 GB; **TDP:** 250 W.

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